Rylie Horning

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Education

- **Virginia Tech** — Blacksburg, Virginia
  B.S. in Electrical Engineering - GPA: 3.72/4.00 — *August 2024 - May 2028*

Skills

- **Electrical Design:** Cadence OrCAD/Allegro, Altium Designer, Circuit Design, PCB Design, LTspice
- **Programming:** Verilog, VHDL, Python, C/C++
- **Embedded Systems:** Arduino, ESP32, Raspberry Pi
- **Test Equipment:** Oscilloscopes, Function Generators, Digital Multimeters, Power Supplies

Experience

- **Electrical Engineering Intern** — East Aurora, NY / Remote
  Moog Inc. — *May 2026 - Present*
  - Designed a custom soldering practice kit from concept through PCB design to support onboarding of technicians.
  - Developed circuit schematics and PCB layout using Cadence OrCAD/Allegro.
  - Currently testing and refining the initial design to improve functionality and inform the next PCB revision.

- **Powertrain Subteam Member** — Blacksburg, VA
  BOLT Virginia Tech — *January 2026 - Present*
  - Designing electrical systems for our second fully-electric dirt bike.
  - Redesigning the low-voltage power architecture to operate from the main battery pack rather than an auxiliary battery.
  - Designing power distribution PCBs for the dirt bike using Altium Designer.

- **Digital Systems Teaching Assistant** — Blacksburg, VA
  Virginia Tech ECE Department — *August 2026 - Present*
  - Assist students with digital logic design, Verilog implementation, and FPGA development.
  - Troubleshoot student designs and explain hardware debugging techniques.

- **Supply Chain Intern** — Christiansburg, VA
  Moog Inc. — *July 2025 - May 2026*
  - Created work instructions using Microsoft Word and ensured documents met the ISO 9001 standard.
  - Used SAP to support internal workflow and reporting tasks as needed.

- **Electrical Engineering Intern** — Blacksburg, VA
  Virginia Tech Electric Service — *October 2024 - July 2025*
  - Revised campus power distribution schematics and facility electrical maps using AutoCAD Electrical.
  - Gathered equipment data using a Trimble GPS device for mapping purposes.

- **Electrical Engineering Intern** — Radford, VA
  Motion Control Systems Inc. — *January 2024 - September 2024*
  - Conducted testing on motor control PCBs across varying voltages to evaluate performance and thermal behavior.

Projects

- **Home Audio System** — Analog Circuit Design, LTspice, Active Filters, Class-D Amplifier, Arduino, PWM
  - Designed and integrated a multi-stage audio system with analog filtering, Class-D amplification, and microcontroller interfacing.
  - Verified performance and behavior through LTspice simulation and hardware validation.
  - Implemented MOSFET switching stage driven by Arduino-generated carrier (>80 kHz) and designed a Butterworth RLC low-pass filter to meet headphone output specifications.

- **Power Supply Failure Detection System** - Analog design, test and validation
  - Designed and tested an analog voltage monitoring circuit to detect out-of-range conditions.
  - Simulated edge cases in LTspice and compared results with physical oscilloscope measurements.
  - Implemented comparator and BJT-based switching for fault indication.